

Acting in Parallel Without Talking: Focal Points, Collisions, and Length Signals in Diffusion-LM Agents and LLM Agent Teams

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“Can you check my loan eligibility for a home loan of amount \$500,000 from HSBC with annual income \$100,000 and for Wells Fargo for a amount of \$700,000 with annual income of \$120,000?”

Slot lengths	Call 1: institution, amount, income	Call 2: institution, amount, income	Verdict	What went wrong
exact	HSBC, 500000, 100000	Wells Fargo, 700000, 120000	correct	—
one extra mask	HSBC Bank , 5000000, 1000000	Wells Fargo Bank , 7000000, 1200000	wrong	every value is lengthened; amounts $\times 10$
swapped	Wells Fargo , 500000, 100000	HSBC , 700000, 120000	wrong	the banks are bound to the wrong calls

Figure 1: A dLLM agent’s parallel turn under three slot lengths (a verbatim BFCL request). The interface writes the syntax in advance (a *skeleton*) and leaves each value as a *slot* of ℓ mask tokens, its slot length (Figure 2a). Dream, a 7B diffusion LLM, fills the slots *one token per step*, so no two decisions are ever made at once. Wrong parts in bold red. With one extra mask per slot, every value grows; with the two institutions’ slot lengths swapped, each call takes the bank whose name fits its slot: an error that looks like a coordination failure of parallel decoding, made without any parallel decoding.

Abstract

When an agent issues several tool calls at once, or a team of large language model (LLM) agents splits a task, each decision must take its own role without seeing the others. We study where such roles come from, with diffusion LLMs (dLLMs) as a test bed: a dLLM agent fills the value slots of a fixed call skeleton, and its decoding schedule sets which slots decide together and what each sees. We compare these slots with teams of 7B LLM agents that each make one call. On the parallel calls of the Berkeley Function Calling Leaderboard, slots that decide in the same step take the entities in the order the request mentions them: committing 16 tokens per step instead of one costs the dLLM Dream 5.3 points of set accuracy. Simultaneous Dream agents that get their role as a number in the prompt mostly fail (28.8% set accuracy, against 81.7% for the slots; 48.1% when told the rule), while the same agents each filling their own call of the whole skeleton reach 85.6%; agents that take turns come within 4 points of one agent making all calls. When nothing orders the choices, simultaneous agents collide, and sampling does not prevent it. Slot length is a strong cue: wrong lengths corrupt values and bind them to the wrong calls even with one token per step; a pre-written closing delimiter and a type mask, which we isolate, account for part of this. These errors look like failures of parallel decoding, so evaluations of skeleton-based dLLM agents should control slot lengths before blaming parallelism.

Keywords

LLM agents, multi-agent LLM systems, tool use, parallel action selection, role assignment, coordination without communication, focal points, diffusion language models

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1 Introduction

Agents increasingly act in parallel. A tool-calling agent answers one request with several calls at once, for example to get the weather in two cities or to check a loan at two banks, and in multi-agent large language model (LLM) systems several workers divide one task among themselves concurrently [27, 37]. Whenever their view of each other is delayed or absent, each parallel decision must take its own role, such as an entity or a subtask, without seeing what the others take. If two decisions take the same role, an action is duplicated; if they swap roles, actions are bound to the wrong objects. Agents that cannot communicate have known ways to coordinate: a focal point that all of them notice [12, 39], a shared convention or social law [22, 41], labels that break the symmetry, or taking turns. We ask which of these work for LLM-based deciders.

Diffusion large language models (dLLMs) make this question precise. A dLLM starts from a canvas of mask tokens and, at each denoising step, commits some of them; tokens committed in the same step are chosen without seeing each other. A dLLM agent can write all the calls of a turn on one canvas, so its decoding schedule decides which parts of the turn are chosen together and what each part sees, and we can vary it exactly. dLLM agents also matter in their own right: they often fail at tool calling, and the failures have been blamed on diffusion noise and parallel decoding [26]. To keep every call well formed, such an agent can write the syntax of its turn in advance as a skeleton and leave only the argument values masked. Each value then gets a slot with a fixed number of masks, set before the value is known. Figure 1 shows what this does even when the agent commits one token per step. With one extra mask per slot, HSBC becomes “HSBC Bank” and a \$500,000 loan becomes



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one of \$5,000,000; with the slot lengths of the two banks swapped, the banks change calls.

We treat the slots of such a turn as deciders that share one policy and differ only in their position and in what they observe, and we describe who decides together and who sees what in the terms of team decision problems [36]. Three cues could set the deciders apart: the order in which the request mentions its entities, each decider’s role (a position on the shared skeleton, or a number in an agent’s prompt), and the slot length. We test hypotheses about each on the parallel calls of the Berkeley Function Calling Leaderboard (BFCL) [33] and on symmetric requests that leave the assignment open, with two dLLMs and an autoregressive (AR) model. We give the same requests to teams of 7B LLM agents that each make one call, at the same time or in turns, and anonymous, numbered, or told the rule that agent i makes the i -th mentioned call.

We find three things. (1) When the request orders its entities, slots that decide in the same step take them in that order: committing 16 tokens per step instead of one costs Dream only 5.3 points of set accuracy (the share of requests whose calls are all correct). Simultaneous agents that get their role as a number in the prompt mostly fail (numbered Dream agents: 28.8%, or 48.1% when told the rule, against 81.7% for the slots on the same requests), even where the whole skeleton tells a slot no more than its number; the same agents filling their own call of the whole skeleton, without seeing the others’ choices, reach 85.6%. Agents that take turns come within 4 points of one agent that makes all calls. (2) When nothing orders the choices, simultaneous agents collide whether numbered, told the rule, or sampled; turn-taking prevents the collisions, and so does a confidence threshold. (3) Slot length is a strong cue: surplus masks lengthen values, and a slot with a sibling’s length often takes that sibling’s value, even with one token per step, partly through two interface choices we isolate, a pre-written closing delimiter and a type mask; removing either recovers part of the loss. With the model’s own length estimates, one token per step produces more cross-call errors than 16 tokens per step with exact lengths.

Our contributions are: (i) a test bed that separates the protocol, the role cues, and the action interface of parallel LLM deciders (Section 3); (ii) evidence that, on ordered requests, a position on a shared skeleton guides parallel deciders, slots and Dream agents alike, while a number or a rule in a 7B agent’s prompt mostly does not, and that turn-taking prevents collisions on symmetric requests (Sections 4 and 5); (iii) an analysis of slot length as a cue, with controls for the closing delimiter and the type mask (Section 6); and (iv) an attribution test and design guidance for skeleton-based dLLM agents (Sections 7 and 8). Length biases of dLLMs are known for single spans [16, 25, 47]; we show what they do to the joint actions of an agent.

2 Related Work

Tool-calling agents and parallel generation. LLM agents interleave reasoning with tool calls [40, 50], and many requests need several calls in one turn. LLMCompiler [20] plans such calls and executes them in parallel, BFCL [33] evaluates parallel calls in dedicated categories, and Skeleton-of-Thought [32] has an LLM write the skeleton of an answer and expand its points in parallel. These

works parallelize execution or expansion; we study how one model generates a turn’s calls in parallel, and how teams split them.

dLLM agents. Masked diffusion language models [8, 31, 38, 51] are being tried as agent backbones. Lu et al. [26] find that dLLM agents fail at tool calling, often with malformed JSON or wrong parameters, and attribute this to diffusion noise and to parallel decoding that weakens causal dependencies. They vary neither slot lengths nor the tokens committed per step. With matched fine-tuning, dLLM agents run over 30% faster than AR agents but are more prone to structured tool-call failures [53]. Other work overlaps reasoning with tool execution [52], traces retry loops to confidence-ordered commits [34], or steers a frozen dLLM with AR agents [19].

Coordination without communication. Agents that cannot communicate can still cooperate by reasoning about each other’s decision procedures [13], use focal points as a coordination heuristic [12, 21], or follow conventions designed off-line as social laws [41]. Human focal points have been explained by salience [29] and by team reasoning over the labels of the options [44], which experiments support in one setting and cognitive hierarchy theory in another [5]; in dispersion games the players must instead spread over the actions [15]. Zero-shot and label-free coordination ask which conventions independently trained partners can share [17, 45], and ad hoc teamwork asks agents to cooperate with unfamiliar teammates without pre-coordination [42]. Identical anonymous processes in a symmetric network cannot break symmetry deterministically [3], a simulated robot soccer team maps agents to positions of a pre-agreed formation [43], and Dec-POMDPs formalize who sees what [7]. In multi-agent reinforcement learning, sharing parameters across agents with different roles can hurt [10]; our deciders share one policy and get their roles only through their inputs. LLMs often find focal points without communicating [2], coordinate less well when they must model their partners [1], and choose alike by default [4]; multi-agent LLM systems fail, among other reasons, through misalignment between agents [9], and parallel workers coordinate through a shared attention cache [37] or a shared context and task queue [27].

Parallel decoding and length in dLLMs. Tokens committed in the same step are sampled without their mutual dependencies [24]. Samplers bound the resulting error with a confidence threshold [46] or an entropy bound [6], and ParallelBench [18] documents invalid combinations such as “New City”. A fixed canvas length is a known limitation: code infilling degrades when the number of masks does not match the completion [47], fixed format anchors truncate content or add redundant content [16], and ample masks for schema fields lead to over-generation [48]. The remedies size a single span or response: by searching its length [25], growing or shrinking the canvas [47], estimating where a span ends [16], or reading end-of-sequence signals [23, 49]. Constrained decoding enforces grammars [14, 30] or finite automata [11]; our skeleton is the simplest such constraint. We do not claim to discover the length bias; we measure it in actions with several slots and show that it binds values to the wrong calls.

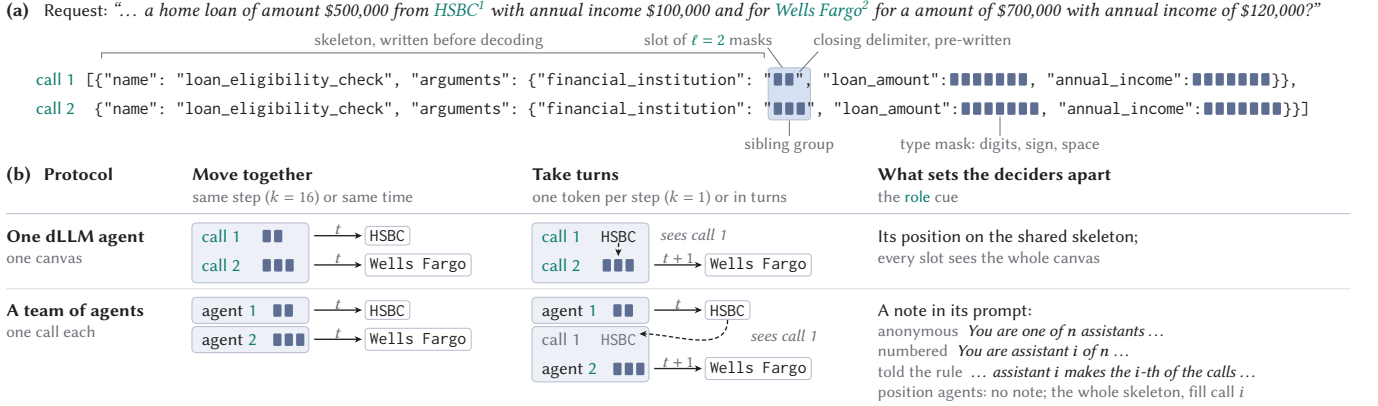


Figure 2: The test bed. (a) The skeleton interface. (b) Deciders that move together do not see each other’s choices; those that take turns see the earlier ones (dashed). Teal: cues that can set deciders apart (Section 3.3).

3 Parallel Deciders, Protocols, and Role Cues

3.1 The Agent and Its Action Interface

A tool-calling agent receives a request q and a set of tool schemas, and acts by issuing a set of calls $A = \{a_1, \dots, a_n\}$, each naming a tool and binding its arguments. When $n > 1$, the turn is a joint action. We score it as a set, as BFCL does [33]: the turn is correct if its calls match the reference calls one to one, in any order.

The dLLM agent writes A as a JSON array onto a canvas of mask tokens (Figure 2a). To make every action well formed, its interface writes the syntax of the turn in advance as a *skeleton* (the number of calls, the tool and argument names, all brackets and quotes) and masks only the argument values, each in a *slot* of ℓ mask tokens, its *slot length*. The slots that hold the same argument in different calls of the same tool form a *sibling group*. Our skeleton is an oracle, built from the reference calls, so that every remaining error is an error in a value; Section 9 discusses what the oracle hides.

Two details of the interface matter below. First, the skeleton writes the closing delimiter of each value after its slot (the closing quote of a string, or the comma or brace after any other value). A value ends early when the model writes its own closing token inside the slot; the rest of the slot is then set to padding, and both are dropped from the output. Second, a *type mask*, as in grammar-constrained decoding, lets each slot accept only tokens of its argument’s JSON type, its closing tokens, and padding: string slots exclude tokens with a quote or backslash, integer slots accept only digits, a sign, and a leading space, and float slots also a point and an exponent. Section 6 varies both. The AR agent uses the same skeleton and type mask, stopping at its closing token or the slot length.

3.2 Deciders and Protocols

At each denoising step, the dLLM predicts every masked position from the current canvas and commits some of them, never revising a committed token; tokens committed in the same step are chosen given the canvas *before* that step, so they cannot see each other. We call each slot a *decider*, since its first content token usually decides which entity it names. The deciders are positions of one network:

they share one policy and differ only in their position on the canvas and in what has been committed when they move. In the terms of team decision problems [28, 36], a parallel turn has one payoff (the turn is correct or not) and an information structure that the decoding schedule sets. Unlike in team theory, the policy is fixed by training and never optimized for this payoff; we use the terms only for who moves together and who sees what, and the predictions below are hypotheses, not consequences of the model.

To compare, we give the same requests to *teams of n LLM agents*, one per call. Each agent is a separate generation of the same model (Qwen2.5, or Dream decoding one token per step in confidence order), sees the skeleton of its own call only, and receives the request followed by a note on the protocol (Figure 2b; the supplementary material gives all texts). A *protocol* says which deciders move together and what each sees:

- *One canvas*. With k tokens per step in confidence order, the k most confident masked positions move together; with $k = 1$, *one token per step*, every decider sees every earlier commitment. Left-to-right decoding (LTR) fixes the order of moves. A confidence threshold $\tau = 0.9$ [46] commits every position whose top probability exceeds τ , and the most confident position alone when none does.
- *Teams*. Agents act at the same time, seeing no other call, or in turns, seeing the calls made so far. They are anonymous, numbered, or numbered and told the rule (“By convention, assistant i makes the i -th of the calls, in the order in which the request mentions them”). *Position agents* get the whole turn’s skeleton instead of a note, and fill call i only; the other calls’ values stay masked (Dream) or are placeholders (Qwen2.5, which reads only the earlier calls), so an agent sees where its call stands but none of the other agents’ choices.

3.3 Role Cues and Hypotheses

The sibling deciders of a turn must each take a distinct entity, the one its own call needs. A shared policy can set them apart only through something they share that orders the candidates, or something that differs between them:

- (1) *Mention order*. The request names its entities in some order. Taking the i -th mentioned entity for the i -th call is a convention any decider knowing its index can follow; on set-scored requests every assignment is correct, and a shared policy is most likely to reproduce the mention order.
- (2) *Role*. A role is a slot’s position on the shared skeleton, or an agent’s number in its prompt. A role sets deciders apart only if the policy’s output depends on it, and helps only if something orders the candidates: “you are the second call” does not say which of all US cities is the second.
- (3) *Slot length*. In training every mask hides exactly one token, so the mask count always equals the length of what it hides. An interface that sets slot lengths supplies this cue whether or not it is correct.

We test three hypotheses about these cues (Sections 4–6).

- P1** *Order coordinates*. When the request orders its entities, slots that decide in the same step take the i -th entity for the i -th call, so committing many tokens per step costs little.
- P2** *Symmetric deciders collide*. When nothing orders the candidates, deciders that move at the same time take the same value, and deciders that take turns do not.
- P3** *Length is a cue*. A slot’s length steers its value: a slot longer than its value gets a longer value, and a slot with a sibling’s length gets that sibling’s value, at any k , including one token per step.

If a prompted number acts as a role, simultaneous numbered agents take the i -th entity of an ordered request or list without colliding.

3.4 Models, Data, and Measures

Models. Dream [51] (Dream-v0-Instruct-7B) is a dLLM with full bidirectional attention, initialized from Qwen2.5-7B. LLaDA2.0 [8] (LLaDA2.0-mini, a mixture of experts with 16B parameters, 1.4B active per token according to its model card) is a dLLM that denoises 32-token blocks left to right; only 3.1% and 2.3% of the sibling slot pairs in the two BFCL categories share a block, so its calls take turns and it serves as a control. The AR reference is Qwen2.5-7B-Instruct [35], the instruction-tuned version of the model that Dream is initialized from. All models see the same system prompt, which lists the tools as JSON and asks for a JSON array of calls, and decode greedily unless we sample at temperature $T = 0.7$ (Section 5).

Data. We use the `parallel` and `parallel_multiple` categories of BFCL v4 [33]: 400 requests with 2 to 8 calls (2.9 on average) and 3,063 value slots. Subsets: the *equal-length* requests (213 for Dream’s tokenizer, 216 for LLaDA2.0’s), in which each argument has slots of one length in every call that uses it; the 104 *team-symmetric* requests, in which the sibling calls of each tool share one skeleton and slot lengths, so that only an agent’s prompt sets it apart from the agents that call the same tool (86 of them use one tool); the 165 *swap* requests, which have sibling slots of different lengths; and LLaDA2.0’s subset of 100 requests (34 of the swap requests) for its runs with wrong lengths. For requests in which nothing orders the calls, we add 105 *choose-N* requests (Section 5).

Slot lengths. *Exact*: each slot has as many masks as its own reference value has tokens, tokenized on its own. *Surplus +s*: every slot gets $s \in \{1, 2, 4, 8\}$ extra masks. *Swap*: slot lengths are rotated

Table 1: One canvas, ordered requests, exact slot lengths (400 BFCL requests; set accuracy and CCER in %). (eq.): the equal-length requests (213 for Dream, 216 for LLaDA2.0). Order: share of the equal-length parallel requests (84, 87) whose i -th call takes the i -th mentioned entity (outputs that do not parse count as out of order). Fwd: forward passes per request. LTR: left to right; τ : threshold 0.9; –: not run. Qwen2.5 (AR) in the same skeleton: 86.5% set accuracy, 1.0% CCER.

	$k=1$	2	4	8	16	LTR	τ
<i>Dream (full attention)</i>							
Set acc.	89.8	88.0	87.0	86.3	84.5	89.3	89.0
CCER	1.0	1.0	1.3	1.5	2.0	1.0	1.3
Set acc. (eq.)	86.9	85.4	84.0	82.6	81.7	86.4	85.4
Order	94.0	94.0	90.5	88.1	88.1	92.9	91.7
Fwd	28.0	14.2	7.3	3.9	2.2	27.8	1.9
<i>LLaDA2.0 (32-token blocks)</i>							
Set acc.	87.8	87.5	86.5	83.3	79.8	–	–
CCER	2.5	2.3	2.8	3.5	3.8	–	–
Set acc. (eq.)	87.5	88.0	86.1	83.8	80.6	–	–
Order	89.7	90.8	89.7	88.5	86.2	–	–
Fwd	27.8	14.8	8.3	5.1	3.9	–	–

by one position within each sibling group. *One slot longer*: in each sibling group with unequal lengths, the first slot shorter than the longest gets the longest sibling’s length, and all others stay exact. These derive from the reference values: they are diagnostic interventions, not lengths an agent could set. *Estimate*: lengths that the model predicts in one forward pass (Section 8).

Measures. *Set accuracy* is the share of requests that the BFCL checker accepts. The *cross-call error rate* (CCER) is the share of requests with at least one cross-call error, found by an optimal one-to-one matching of predicted and reference calls: a duplicated or missing call, a value that belongs to another call (bound), a string that puts a piece of another call’s value in place of a piece of its own (chimera), or an argument that the reference shares across calls but the prediction does not (shared; one wrong copy is enough). At the slot level we read each slot’s value from its own tokens and classify it as its own reference value, a sibling’s value (*fit* if it has exactly the slot’s length), an *overflow* (its own value followed by more), or anything else. *Order* measures whether the i -th call takes the i -th mentioned entity, with the reference order as a proxy for the mention order (they differ in 9 of the 104 team-symmetric requests). For paired comparisons on the same requests we give the difference with a 95% bootstrap interval over requests and the p -value of an exact McNemar test, and say where a result does not survive Holm’s correction over all our 48 paired tests; for single rates, a Wilson interval.

4 Ordered Requests: Positions Coordinate, Numbers Do Not

We first test **P1** on BFCL with exact slot lengths (Table 1).

Same-step slots coordinate. At 16 tokens per step, Dream commits the first content tokens of 74.9% of its sibling slot pairs in the same step, so most siblings choose their entities without seeing each other.

Table 2: One canvas against teams of agents. Ordered: set accuracy (%) on the 104 team-symmetric BFCL requests. Symmetric (choose-N): Dup. is the share of requests (%) in which two calls take the same city; Order, the share of list requests whose i -th call takes the i -th listed city. Agents: Figure 2b; Dream agents decode one token per step. Greedy decoding on 60 list and 45 open requests; $T=0.7$: sampling at temperature 0.7, five seeds. Open requests hold one-token slots, so only duplicates are reported. Qwen2.5’s position agents copy the placeholder of earlier calls in choose-N (Section 5). –: not run or not applicable.

	Ordered		Symmetric, greedy		$T=0.7$	
	Set		List	Open	List	Open
	acc.	Dup.	Order	Dup.	Dup.	Dup.
<i>One model writes all n calls</i>						
Dream, $k=16$	81.7	15.0	11.7	100.0	17.3	100.0
Dream, $\tau=0.9$	86.5	0.0	48.3	0.0	–	–
Dream, $k=1$	88.5	0.0	48.3	0.0	0.0	0.4
LLaDA2.0, $k=16$	81.7	0.0	96.7	4.4	–	–
LLaDA2.0, $k=1$	88.5	0.0	100.0	0.0	–	–
Qwen2.5 (AR)	91.3	0.0	81.7	0.0	–	–
<i>A team of n Qwen2.5 agents, one call each</i>						
same time, anonymous	0.0	100.0	0.0	100.0	99.7	93.8
same time, numbered	0.0	100.0	0.0	97.8	100.0	93.8
same time, told the rule	34.6	70.0	8.3	–	–	–
turns, anonymous	88.5	0.0	91.7	0.0	–	–
turns, numbered	88.5	0.0	91.7	0.0	–	–
same time, position	54.8	–	–	–	–	–
<i>A team of n Dream agents, one call each</i>						
same time, anonymous	0.0	100.0	0.0	100.0	87.7	100.0
same time, numbered	28.8	78.3	3.3	100.0	75.0	100.0
same time, told the rule	48.1	45.0	10.0	–	–	–
turns, anonymous	84.6	0.0	35.0	0.0	–	–
turns, numbered	85.6	0.0	26.7	0.0	–	–
same time, position	85.6	15.0	11.7	100.0	–	–

This lowers Dream’s set accuracy only from 89.8% to 84.5% (by 5.3 points, 95% interval [3.3, 7.5]) while cutting the forward passes per request by a factor of 13. The confidence threshold commits 93.3% of sibling pairs together and keeps 89.0% with 1.9 forward passes per request ($p = 0.45$). On the parallel requests with equal lengths, where lengths cannot tell the calls apart, the i -th call takes the i -th mentioned entity in 94.0% of requests with one token per step and in 88.1% with 16; among the outputs that parse with every call matched, the share stays between 95.0% and 96.1% at every k . The cost of parallelism lies inside single values: of the 21 requests that Dream gets right with one token per step and wrong with 16, 13 have only single-call errors and 5 no longer parse, against 3 with cross-call errors, and its cross-call error rate rises only from 1.0% to 2.0% ($p = 0.13$). LLaDA2.0, whose calls take turns block by block, loses 8.0 points [5.3, 10.8], also mostly in single values. The slots rarely face a hard choice: most values are copied from the request, and our skeleton lays the calls out in the reference order, which BFCL usually gives in mention order.

Simultaneous agents do not. On the 104 team-symmetric requests, the agents that call the same tool see the same skeleton, so only their prompts can set them apart (Table 2, left). Anonymous agents that act at the same time receive identical inputs and make identical calls, so no team is correct. Numbering raises Dream’s agents to 28.8% and leaves Qwen2.5’s at 0%, against 81.7% for Dream’s canvas at 16 tokens per step on the same requests (difference 52.9 points, [42.3, 62.5]). Told the rule, they reach 48.1% and 34.6%; most numbered agents take the entity mentioned first (67.8% of Dream’s and 88.3% of Qwen2.5’s agents in sibling groups). The team agents see only their own call, but this does not explain the gap: on the 86 requests with one tool, the whole skeleton is n copies of one call’s skeleton, so a slot knows little more than a numbered agent: its index among n identical calls and, for most sibling pairs, not the other’s choice. There the canvas reaches 81.4%, numbered agents 32.6% (difference 48.8 points, [38.4, 59.3]), and agents told the rule 55.8%. Position agents, which see the whole skeleton but none of the other agents’ values and fill their own call, close the gap for Dream: they reach 85.6%, 56.7 points above its numbered agents ([47.1, 66.3]) and 3.8 above its canvas ($p = 0.125$). Qwen2.5’s reach 54.8%, above its agents told the rule (difference 20.2 points, [8.7, 31.7]), although 43 of its 289 agents copy the placeholder of an earlier call into their own. For the same model and the same lack of information about the others, filling one’s own call of the whole skeleton works, and a number in the prompt does not. Turn-taking agents come within 4 points of one agent making all calls: Dream’s teams reach 84.6–85.6% against 88.5% for its canvas with one token per step, and Qwen2.5’s 88.5% against 91.3%; each team loses 3–4 of the one agent’s correct requests and gains at most one ($p \geq 0.125$).

P1 holds for same-step slots and for Dream’s position agents: a position on the shared skeleton acts as a role, and the request’s order assigns the entities to roles. For simultaneous 7B agents, a number or even a stated rule in the prompt is a weak role.

5 Symmetric Requests: Simultaneous Agents Collide

We now take the order away. A *choose-N* request asks for the weather in $n \in \{2, 3, 4\}$ cities without saying which call takes which city, either from a list of $n + 3$ one-token cities (“...any three of these cities: Boston, Dallas, Toronto, Austin, Chicago and Paris”; 60 requests) or openly (“...three different US cities of your choice”; 45 requests). Any n different allowed cities are correct, so the deciders must spread over the candidates, as in a dispersion game [15]. Every city slot holds one token, so each call makes a single decision. At 16 tokens per step, every city slot moves in the first step from the all-masked canvas: each slot sees the whole skeleton but none of the other choices. We test **P2** on the list requests; the open ones serve as illustrations, since greedy decoding gives one to ten distinct outputs per n across their paraphrases, and the one-token slot shortens many names (34.1% of Dream’s cities at one token per step, 74.1% of Qwen2.5’s).

One canvas. Dream’s same-step slots duplicate a city in 15.0% of list requests ($p = 0.004$ against none with one token per step; Holm $p = 0.059$) and 5.5% of pairs, fewer than independent uniform choices would (43.2% of requests: 20.0%, 44.4%, and 65.0% for $n = 2, 3, 4$). Their positions set them apart more often than chance,

though not in list order (11.7% of requests). With one token per step, left to right, or the confidence threshold, no request has a duplicate: under the threshold, two city slots move together in only 8 list requests, where each is confident about a different city, and in every other step at most one slot exceeds 0.9. LLaDA2.0’s calls take turns block by block and rarely collide (2 of its 76 same-block, same-step pairs), and the AR agent never collides.

Teams of agents. Anonymous agents that act at the same time make identical calls in every team, as their inputs are identical. Agents that take turns never collide. Agents that act at the same time collide whether or not they are numbered: numbered Qwen2.5 agents take the same city in every list team and numbered Dream agents in 78.3% (numbering lowers the duplicates by 21.7 points, [11.7, 31.7]). Told the rule, Qwen2.5 and Dream agents still collide in 70.0% and 45.0% of list teams and take the cities in list order in only 8.3% and 10.0%, although the rule’s wording, which refers to the order of the calls, may fit a list of cities poorly. What the agents share is primary salience, the choice each finds salient for itself, rather than a choice made to match the others [5, 29]: with greedy decoding, every anonymous or numbered Qwen2.5 agent that acts at the same time takes the city listed first, and 55.0% of those told the rule do. The canvas of Section 4 fits the same description: its slots follow the mention order because the shared policy prefers that order, and the skeleton lays the calls out in it. Sampling does not break the symmetry: at $T = 0.7$, with a separate seed for every agent, teams that act at the same time collide in 75.0–100% of list runs (99.0–99.8% of Qwen2.5’s agents still take the first city), while Dream’s same-step slots collide in 17.3%. Position agents see all n city slots: Dream’s choose exactly as its canvas at 16 tokens per step does in all 105 requests, since they see the same canvas, and Qwen2.5’s agents after the first mostly copy the placeholder of the earlier calls instead of naming a city (188 of 210).

What carries over. P2 holds for teams of 7B agents: a number in the prompt, a stated rule, or sampling leaves 45–100% of simultaneous list teams colliding, while agents that take turns never collide. For one canvas it holds only in part: a position on the skeleton sets same-step slots apart more often than chance, and the threshold lets symmetric choices move one at a time.

6 Slot Length Steers Values

We now turn to the interface. Sections 4 and 5 gave every slot exactly as many masks as its reference value has tokens, which an agent in deployment cannot do.

One extra mask. With one token per step, one extra mask per slot lowers Dream’s set accuracy from 89.8% to 15.5% and LLaDA2.0’s from 92.0% to 16.0% (on its 100 requests; Figure 3a,b). The share of Dream’s slots that hold their own value falls from 96.0% to 37.5%, and 44.4% hold their own value followed by more (Table 3; Figure 1). Near misses hurt most: set accuracy is lowest with two extra masks (3.3%) and recovers with eight (60.0%; LLaDA2.0: 38.0%). With one or two masks left after the value, Dream gives a closing token only 1.0% and 3.3% of its probability (Figure 3c). It would rather write something else, but often not content: its most probable token at the first surplus position lies outside the slot’s type mask in 59.3% of slots, mostly an escaped line break in strings (1,076 of 1,464) and a

Table 3: Slot length steers values (Dream, one token per step). Share of slots (%) that hold their own value, their own value followed by more (overflow), the value of a sibling whose length fits the slot, or anything else: over all 3,063 slots (exact, +1), the 514 slots whose length the swap changes, and the 243 slots lengthened to a sibling’s length while the sibling keeps its exact slot (one slot longer). Each slot’s value is read from its own tokens. Set accuracy (%) on the 400 requests, or on the 165 whose slots the swap changes. Agents are anonymous; for them, overflow counts as other (–).

	Own	Over- fill	Sibling that fits	Other	Set acc.
<i>One canvas, closing token after the slot</i>					
exact	96.0	0.2	0.9	3.0	89.8
+1	37.5	44.4	0.2	17.9	15.5
swap	5.4	2.3	79.4	12.8	67.3
one slot longer	21.8	20.2	35.4	22.6	17.6
<i>One canvas, closing token in the slot</i>					
exact	89.9	4.8	0.8	4.5	70.3
+1	62.5	26.2	0.1	11.2	26.8
swap	24.7	9.3	49.2	16.7	35.8
one slot longer	40.3	23.9	14.0	21.8	24.8
<i>One Dream agent per call (the slots the swap changes)</i>					
same time, exact	85.8	–	4.9	9.3	48.5
same time, swap	1.9	–	89.1	8.9	46.7
turns, swap	2.3	–	84.2	13.4	75.8

type suffix after numbers; with two extra masks, 965 of 1,014 integer slots would write a decimal point (\$500,000.0). The type mask turns this filler into content. Format-tolerant slots, which also accept such tokens and remove them when decoding, raise set accuracy with one and two extra masks to 28.3% and 20.0% (differences 12.8 and 16.8 points, both $p < 10^{-4}$), and the share of slots that hold their own value with one extra mask from 37.5% to 56.7%, while leaving exact lengths almost unchanged (90.0% against 89.8%). Still, 24.8% of the tolerant slots hold their own value followed by more content: the type mask explains a part of the damage, and the rest comes from the closing delimiter and the mask count.

The closing delimiter. Ending a value early also means writing a closing token right before the one the skeleton has already written, a pattern rare in training. To separate this from the mask count, we move the closing token into the slot: the skeleton no longer writes it, the slot gets one more position, and after the model’s closing token the slot is filled with spaces, which leaves ordinary JSON (Table 3, middle; dashed in Figure 3). For strings the delimiter explains almost all of the effect: with one extra mask, a string closes with probability 69.9% instead of 0.5%. Numbers still grow: an integer closes with 19.3% at one extra mask, against 80.8% when only the closing position is left. Overall, the share of slots that hold their own value falls from 89.9% to 62.5% instead of from 96.0% to 37.5%, and 26.2% still overflow. The variant is no remedy as it stands: 23.3% of its outputs do not parse even with exact lengths (set accuracy 70.3%), so we read slots one by one.

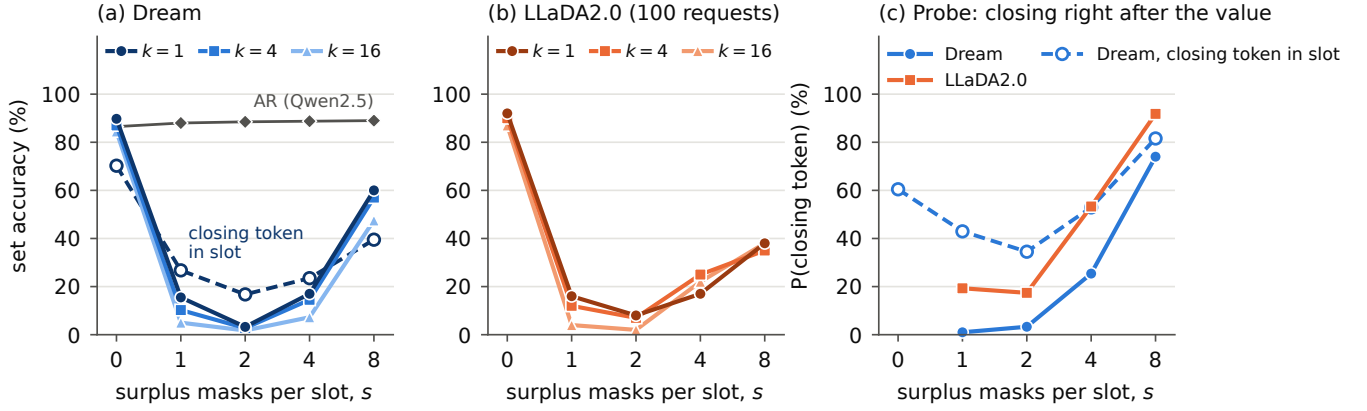


Figure 3: Every slot gets s extra masks. (a, b) Set accuracy against s , one line per k ; LLaDA2.0 on its 100 requests. Dashed: Dream at $k=1$ with the closing token in the slot. Gray: the AR agent, which stops at its closing token. (c) Probe over all 3,063 slots: with the reference value written into the slot (teacher forcing) and s masks left, the mean probability that the next token closes the value; dashed: the closing token in the slot ($1+s$ positions left).

Swapped lengths and one slot longer. Rotating the slot lengths within each sibling group gives every changed slot the length of a sibling’s value. A slot made shorter can rarely hold its own value (5.1% do, through a shorter accepted form); a slot made longer can, by closing early. With one token per step, Dream’s lengthened slots take the sibling value that fits in 79.1% of cases (95% interval over requests [71.8, 85.7]) and keep their own in 5.8%; with the closing token in the slot, 49.2% [41.4, 57.0] and 15.5%. In a swap, though, the sibling also gets a slot of the wrong length and may take the lengthened slot’s value first. When only one slot of a group is lengthened and its sibling keeps an exact slot, the lengthened slot takes the sibling’s value in 35.4% of cases and keeps its own in 21.8%; with the closing token in the slot, the order reverses (14.0% and 40.3%). So a sibling’s length pulls a slot toward the sibling’s value mainly when the delimiter makes closing early unusual, or when the sibling has itself been moved. The type mask plays no part here: with format-tolerant slots, the canvas is identical in 164 of the 165 swap requests.

Agents that make one call. The cue needs no shared canvas. On the 165 swap requests, exact lengths, taken from the reference values, tell each agent which value is its own: anonymous Dream agents that act at the same time reach 48.5% set accuracy (numbered ones 70.9%), anonymous Qwen2.5 agents, which do not see the mask count, 0.6% (difference 47.9 points, [40.0, 55.8]). With swapped lengths, 89.1% of the swapped slots of Dream agents take the sibling value that fits (84.2% when they take turns; Table 3).

P3 holds in part: surplus masks are filled rather than left empty, and a slot with a sibling’s length often takes its value; the delimiter and the type mask each explain part of the damage.

7 Length Errors Masquerade as Coordination Failures

If parallel decoding caused an agent’s errors, we would expect particular ones: a value bound to the wrong call, a duplicated call,

Table 4: What an evaluator sees. Requests (%) with each kind of cross-call error, when the agent commits 16 tokens per step with exact slot lengths (shaded) and when it commits one token per step with wrong slot lengths (for estimated lengths also 16), on the same requests. Estimate: lengths estimated by the model (Section 8). Req.: requests. Dup.: a duplicated call; Bound: a value that belongs to another call; Chim.: a string that puts a piece of another call’s value in place of a piece of its own; Shared: an argument that the reference shares across calls and the prediction does not. A request can have several kinds, so CCER is not their sum.

Slot lengths	Req.	Dup.	Bound	Chim.	Shared	CCER
<i>Dream</i>						
exact, $k=16$	400	0.5	0.8	0.0	0.8	2.0
estimate, $k=1$	400	1.5	3.5	0.0	2.3	6.5
estimate, $k=16$	400	3.0	4.3	0.3	3.8	10.3
+1, $k=1$	400	0.5	0.8	0.0	1.5	2.8
+8, $k=1$	400	1.0	0.5	0.0	0.3	1.8
exact, $k=16$	165	0.0	0.6	0.0	0.6	1.2
swap, $k=1$	165	0.0	15.2	0.0	0.0	15.2
<i>LLaDA2.0</i>						
exact, $k=16$	100	1.0	1.0	0.0	1.0	3.0
estimate, $k=1$	100	0.0	6.0	0.0	7.0	12.0
+1, $k=1$	100	2.0	1.0	1.0	4.0	7.0
+8, $k=1$	100	11.0	26.0	0.0	3.0	35.0
exact, $k=16$	34	2.9	2.9	0.0	2.9	8.8
swap, $k=1$	34	0.0	29.4	2.9	0.0	29.4

a value spliced from two calls. Wrong slot lengths produce the first two even with one token per step (Table 4).

Swapped lengths. With swapped lengths and one token per step, Dream has cross-call errors in 15.2% of the swap requests, against 1.2% at 16 tokens per step with exact lengths (Holm $p < 10^{-4}$), all of them values bound to the wrong call. The rotated sibling group

follows the length while the other arguments of the call follow the order, so one call’s arguments name different entities (Figure 1).

What an evaluator sees. With the lengths that the model estimates itself (Section 8), the condition closest to deployment, Dream’s CCER is 6.5% with one token per step and 10.3% with 16, against 1.0% and 2.0% with exact lengths. Both factors matter: parallel commits add 1.0 point with exact lengths but 3.8 with estimated ones ($p = 0.0026$; interaction 2.8 points, exact sign test $p = 0.052$). Still, one token per step with estimated lengths has more cross-call errors than 16 tokens per step with exact lengths (difference 4.5 points, [2.0, 7.2], Holm $p = 0.018$; without the Shared label, 4.5% against 1.3%, $p = 0.007$; LLaDA2.0: 12.0% against 3.0%, not significant after correction). Surplus masks mostly corrupt single values (Dream’s CCER with one extra mask: 2.8%, not significantly above 2.0%), though with eight, LLaDA2.0 binds a value to the wrong call in 26.0% of its requests. Chimeras are rare. With one token per step, Dream writes none in the conditions of Table 4 and 5 in the others, mostly garbled dates such as “202021-01-01”; its clearest chimeras, such as “San Alto” from San Francisco and Palo Alto, appear at 4 and 16 tokens per step with wrong lengths, where wrong lengths and parallel commits act together.

An attribution test. These results concern skeleton interfaces; we do not claim that the errors reported for dLLMs elsewhere [18, 26] come from slot lengths. But in a skeleton interface, an error that persists with one token per step is not an error of parallel decoding. Before attributing a dLLM agent’s cross-call errors to parallel commits, an evaluator should decode the same requests with one token per step under the agent’s own length policy and report the difference as the cost of parallelism in that interface, and the errors that remain separately.

8 Designing and Evaluating dLLM Agents

An agent in deployment must give every slot a length before it knows the value; we tested two ways to cope.

A bias toward closing. Adding β to the logits of the closing and padding tokens rescues large surpluses (eight extra masks, $k = 4$: from 57.0% to 72.3% at $\beta = 2$, against 87.0% with exact lengths) but not near misses (two extra masks: from 2.8% to 13.8% at $\beta = 8$). We chose β on the evaluation requests, so 72.3% is optimistic.

Lengths from the model. We give every slot as many masks as its type allows, run one forward pass, read the probability of a closing token at each position as the probability that the value ends there if it has not ended before, and take the most likely length. Our estimate is exact for 51.6% of Dream’s slots and 72.8% of LLaDA2.0’s, but all of a request’s slots are exact in only 14.5% and 27.0% of the requests. With these lengths and one token per step, Dream reaches 42.3% set accuracy and LLaDA2.0 39.0%, against 89.8% and 92.0% with exact lengths. Dream’s wrong estimates are mostly too long (91.5%, by a median of 7 tokens), where a closing bias could help (we did not combine the two); LLaDA2.0’s are as often too short, which no bias repairs.

Recommendations. (1) *Interface.* Avoid near misses: a slot one or two tokens too long is the worst choice, and a slot too short rarely holds its value. Let the model write each value’s closing token, so

that strings can end early, and let slots accept the filler and format tokens a model writes into surplus masks, removing them when decoding (with one extra mask, 28.3% set accuracy instead of 15.5%; with the model’s own lengths, 47.5% instead of 42.3%). (2) *Protocol.* A confidence threshold lets ordered calls move together (89.0% with 1.9 forward passes per request) and symmetric calls one at a time, because two symmetric choices are rarely confident at once (Section 5). (3) *Evaluation.* Measure the cost of parallelism under the agent’s own lengths, as in Section 7.

9 Discussion, Limitations, and Conclusion

Beyond one agent. The protocols and role cues behind **P1** and **P2** do not depend on diffusion; the length cue behind **P3** does, since an AR agent does not see the mask count. Our teams used 7B models, one prompt wording per protocol, and no orchestrator. Systems that give their workers a shared task queue, skeleton or attention cache [27, 32, 37] resemble our canvas and position agents more than our numbered teams; we did not test them, or larger models.

Limitations. The oracle skeleton fixes the number of calls, their tools, their argument names, and their order, so our numbers isolate the values and do not describe a deployed agent. Without the skeleton, Dream reaches 75.3% set accuracy at $k = 2$; 12.5% of its outputs do not parse, and its cross-call errors (5.5% of requests) are mostly omitted, extra, or duplicated calls, which the oracle rules out. Exact lengths fit the first acceptable reference value (“IL”, not “Illinois”), and the skeleton also fills optional arguments (10.3% of slots): lifting the AR agent’s cut-off with eight extra masks changes its output in 44 of 400 requests and raises its set accuracy from 86.5% to 89.0%. We tested two dLLMs, which differ in architecture, size, and tokenizer, with greedy decoding except on choose-N. LLaDA2.0 ran its wrong-length conditions on 100 requests (34 for the swap), where a rate near 40% has a 95% interval of about ± 10 points (± 16 on 34 requests), only one of its cross-call contrasts (eight extra masks) survives Holm’s correction, and it did not use its native tool-calling format. The open choose-N requests allow only one-token city names. We did not run placeholders in the style of S^3 [48], a length search [25], or a variable-length dLLM [47].

Conclusion. Parallel deciders that cannot see each other coordinate on what their shared structure gives them: same-step slots, and Dream agents that see their position on the whole skeleton, follow the request’s order through their positions, while simultaneous 7B agents make little use of a number or a rule in their prompts; taking turns works for slots and agents alike. The structure also sends cues no one intended: wrong slot lengths corrupt values and bind them to the wrong calls even with one token per step, errors that look like failures of parallel decoding. Designers should give parallel deciders their roles through structure and get slot lengths, delimiters, and type masks right; evaluators should control slot lengths before they blame parallelism.

Use of AI Tools

We used Claude Opus 5.5 (Anthropic) to help design experiments, write code, and draft and revise the text; the supplementary material describes this use in more detail. The authors checked all results and text and take full responsibility for them.

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